

INF264 - Homework 4

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1 Polynomial kernel SVM

In this exercise, we will learn how to use a polynomial kernel SVM classifier to fit non-linear data.

1. Load and visualize the dataset contained in the file `svm_data_3.csv`. Can you hope to learn anything reasonable with a linear SVM classifier?
2. You can create a polynomial kernel SVM classifier with the class `sklearn.svm.SVC`, where the argument `kernel` is set to `"poly"`. Four parameters are particularly important for this model: `C`, `degree`, `gamma` and `coef0`. Perform a model selection on the aforementioned parameters of the polynomial kernel SVM classifier
3. Evaluate the performance of the best model.
4. In a 2D plot, display the training data, the support vectors, the decision boundaries, the decision hyperplane and the separating hyperplanes of the best model. **Hint:** Since the data features are two-dimensional, hyperplanes are simply lines. You can display these lines by plotting the levels $\{-1, 0, 1\}$ of the `decision_function` method of your model applied to a grid, using the `matplotlib.pyplot.contour` function.
5. In a 3D plot, display the decision surface and the decision values of the training data using the `matplotlib.pyplot.plot_surface` and `matplotlib.pyplot.scatter3D` functions, respectively. **Hint:** The three dimensions correspond to the two features and the decision values returned by the `decision_function` method.
6. How different are the decision hyperplane and separating hyperplanes (from question 3) compared to what you obtained with the linear SVM classifier on linearly separable data? Explain why.
7. Implement your own polynomial kernel as described in <https://scikit-learn.org/stable/modules/svm.html#svm-kernels>.
8. Implement your own decision function and predict function as described in the slides of the course, and compare them to the `decision_function` and `predict` methods of a polynomial kernel SVM classifier from sklearn.